BIOMEDICAL CHROMATOGRAPHY
Biomed. Chromatogr. **21**: 701–707 (2007)
Published online 16 March 2007 in Wiley InterScience
(www.interscience.wiley.com) DOI: 10.1002/bmc.808

ORIGINAL RESEARCH

Simultaneous determination and pharmacokinetic studies of aloe emodin and chrysophanol in rats after oral administration of Da-Cheng-Qi decoction by high-performance liquid chromatography

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Received 2 December 2006; revised 10 January 2007; accepted 11 January 2007

ABSTRACT: A validated high-performance liquid chromatography (HPLC) method was developed for simultaneous determination and pharmacokinetic study of aloe emodin and chrysophanol in rats. It was performed on a reverse-phase C_{18} column and a mobile phase made up of methanol and 0.2% acetic acid (83:17, v/v). The ultraviolet detection was 254 nm. 1,8-dihydroxyanthraquinone was used as the internal standard. The assay was linear over the range 28–2800 ng/mL (r^2 = 0.9993) for aloe emodin and 25.6–2560 ng/mL (r^2 = 0.9991) for chrysophanol. The average percentage recoveries of three spiked plasmas were 98.8–104.8% and 97.7–103.2% for aloe emodin and chrysophanol, respectively. Their RSD of intra-day and inter-day precision at concentrations of 56, 280 and 1400 ng/mL for aloe emodin and 51.6, 258 and 1290 ng/mL for chrysophanol were less than 3.5%. This method was applied for the first time to simultaneously determinate aloe emodin and chrysophanol in rats following oral administration of traditional Chinese medicine of Da-Cheng-Qi decoction. The pharmacokinetic parameters showed that chrysophanol was better absorbed with higher concentrations in plasma than aloe emodin did. They both eliminated slowly in male rats. The assay is suitable for identifying the plasma and tissue levels of aloe emodin and chrysophanol in preclinical investigations. Copyright © 2007 John Wiley & Sons, Ltd.

KEYWORDS: Da-Cheng-Qi decoction; aloe emodin; chrysophanol; pharmacokinetics; HPLC; traditional Chinese medicine

INTRODUCTION

Currently, Da-Cheng-Qi-Tang (DCQD, Dai-joki-to) is a well-known and widely used traditional Chinese medicine prescription (TCMPs) in China and East Asia. It was first described in AD 200 in the most prestigious medical treatise of China, *Shan Han Lun*. DCQD was traditionally used as the purgative for the treatment of constipation and for clearing internal heat in the stomach and intestine (Katakai *et al.*, 2002). In the past it was widely used to manage dysentery, cholecystitis, cholelithiasis, cholangitis, ileus and peritonitis. It is also used in postoperative care for the patients after abdominal operations (Tang *et al.*, 2005; Xia *et al.*, 2005). In China, DCQD has been particularly used to treat acute pancreatitis for more than 30 years (Liu *et al.*, 2004; Xia *et al.*, 1999).

The prescription is composed of Dahuang (*Radix et Rhizoma Rhei*), Houpu (*Magnolia officinalis Rehd.*), Zhishi (*Fructus Aurantii Immaturus*) and Mangxiao (Natrii Sulfas), with a crude herbs ratio of 1:1:1:1. Dahuang is the principal drug in DCQD, which exerts the main therapeutic effect of the prescription. DCQD has many potential effective components, including aloe emodin and chrysophanol from Dahuang. The pharmacokinetics of aloe emodin after intraperitoneal administration has been reported (Zaffaroni *et al.*, 2003). In humans, no aloe emodin was detected in the plasma after the administration of Agiolax and Sennatin (Krumbiegel and Schulz, 1993). As far as we know, no pharmacokinetic study of chrysophanol had been reported. Furthermore, the published studies have

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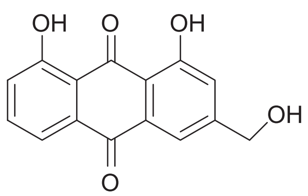
Abbreviations used: DCQD, Da-Cheng-Qi decoction; TCMP, traditional Chinese medicine prescription.

Contract/grant sponsor: National Natural Science Foundation of China; Contract/grant number: 30400576, 90409010.

mainly focused on a single monomer, vegetable drug or dietary drug, but not on complicated prescriptions based on the theory of traditional Chinese medicine and their traditional use. These studies ignored the integration and interacting roles of the herbs in a traditional formula.

With regard to the Chinese herbal formula DCQD, most studies have focused on the pharmacodynamics of the entire prescription (Li and Xiao, 1987). There have been no pharmacokinetic studies or quantitative analyses of the active components in the plasma after traditional administration of DCQD. In detail, there is no information on whether or not the active components are absorbed in the experimental subjects after oral administration. Therefore, it is important to perform preclinical studies of DCQD to determine the absorption of the active components into plasma following traditional administration. Furthermore, it would be of interest to establish a simple and easy HPLC assay for the analytes in order to determine their pharmacokinetic parameters. Additionally, it is practical to optimize the oral dosage schedule of DCQD in clinical practice.

Two compounds (Fig. 1) were selected as the markers to examine the preclinical pharmacokinetics of DCQD. The first one was aloe emodin, a hydroxyanthraquinone of rhubarb that exhibits many activities, such as regulation of gastrointestinal motility, promotion of appetite, a laxative effect on the large intestine and protection against acute liver injury (Huang *et al.*, 2000; Zhao and Liu, 2003). Scavenging of superoxide anion free radicals and hydroxy free radicals has also been reported (Arosio *et al.*, 2000). Another selected compound was chrysophanol, one of the common anthraquinones in Dahuang (Mueller *et al.*, 1998). These two compounds from DCQD exert the main laxative effect of the prescription; thus, they were judged to be suitable indicators for the pharmacokinetic study of absorbed active components after oral administration of DCQD.



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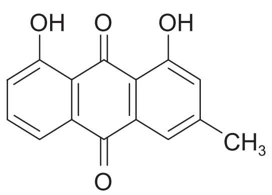

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Figure 1. Chemical structures of aloe emodin (**1**) and chrysophanol (**2**).

This study aimed to explore whether or not these three active compounds of rhubarb would be absorbed into the body. If they were, the study aimed to explore the pharmacokinetics of the absorbed components after oral administration of DCQD in rats by HPLC.

EXPERIMENTAL

Materials and reagents. Dahuang, Houpu, Zhishi and Mangxiao were purchased from Chengdu Green Herbal Pharmaceutical Co. Ltd (Cheng, China). The crude drugs of the formula were extracted twice by refluxing with boiling distilled water (1:12, g/mL) for 1 h, and the solution obtained was concentrated and spray-dried. The dried powder was stored at 4°C until use.

The reference standards of aloe emodin and chrysophanol together with internal standard (IS), 1,8-dihydroxyanthraquinone, were purchased from National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). Methanol was HPLC-grade (Tedia Company Inc, USA). Acetic acid was obtained from Chongqing Chemistry Co. Ltd (Chongqing, China). Water was triple-distilled and was filtered with 0.45 μm Millipore membranes.

Chromatographic conditions. A Waters 2996 liquid chromatograph system consisting of photodiode array detector, a Waters 2695 separation module with an autosampler and an on-line degasser coupled with an Empower chromatography analytical workstation was used. The analytical column was an Inertsil ODS-A C_{18} reversed-phase column (5 μm, 150 × 4.6 mm) with an RP_{18} (5 μm) guard column (both from Dikma). The mobile phase was methanol–0.2% acetic acid (83:17, v/v, pH 3.12) with a flow rate of 1 mL/min. Its detection wavelength was 254 nm. To balance the system, a 10 min lag was maintained between each sampling. The chromatographic separation was performed at room temperature.

Content of aloe emodin and chrysophanol in DCQD. To calculate the administered dose of aloe emodin, rhein and chrysophanol, their contents in DCQD were first quantitatively analyzed. The spray-dried DCQD was mixed with distilled water and diluted to 6 mg/mL. The mixture was centrifuged at 3000 rpm for 10 min and the supernatant solution was obtained, and then 10 μL of this solution were injected into the HPLC system for analysis. The contents of aloe emodin and chrysophanol in the spray-dried extract of DCQD were determined to be 1.71 and 0.55%, respectively, using an internal standard method.

Animals, drug administration and plasma collection. Seven male Sprague–Dawley rats (body weight, 340 ± 25 g) were used, which were born, housed, fed and handled according to the university guidelines and the Animal Ethics Committee guidelines in the animal facility of West China Hospital. The rats were maintained in air-conditioned animal

DOI: 10.1002/bmc

quarters under the following conditions: temperature 22 ± 2°C, relative humidity 65 ± 10%, free access to water, and feeding with laboratory rodent chow (Chengdu, China). The animals were acclimatized to the facilities for 10 days and were then fasted with free access to water for 24 h prior to the experiment.

The spray-dried DCQD was reconstituted with water at a concentration for the crude drug of 2 g/mL. It was orally administered to rats at a dosage of 40 g/kg of DCQD. Blood samples (0.5 mL) were collected through the tail vein at time points of 0 (prior to administration), 0.0833, 0.25, 0.5, 1, 2, 3, 4, 6, 8, 12 and 24 h after a single dose. The rats had free access to water during the experiment. The blood samples were immediately heparinized and centrifuged at 3000 rpm for 10 min, and the supernatant was harvested into 0.2 mL aliquots and stored in 1 mL polypropylene tubes at –20°C prior to analysis. The voucher specimen of the prescription for this study is an aliquot of the same batch; its code number is 20060625, and it has been deposited in the store room of our laboratory.

Extraction procedure. All the standards and the spiked plasma were similarly treated. An aliquot of 0.2 mL plasma and 5 μL of IS (100 μg/mL) were placed in a 1.5 mL polypropylene centrifuge tube and thoroughly homogenized by vortexing, mixed with 80 μL of 2 M hydrochloric acid, and vortexed for 1 min. The analytes were extracted by the addition of 0.8 mL diethyl ether, vortexed for 3 min, and centrifuged at 8000 rpm for 10 min to achieve separation. The supernatant was decanted into a fresh 1.5 mL centrifuge tube and dried under nitrogen gas. The dried residue was reconstituted in 200 μL of methanol, and 40 μL of this solution was injected into the HPLC system.

Calibration procedure. Stock solutions of aloe emodin, chrysophanol and IS (1,8-dihydroxyanthraquinone) were prepared by dissolving each of the substances in methanol. The calibration curves were constructed based on analysis of various concentrations of aloe emodin (28, 56, 140, 280, 560, 1400 and 2800 ng/mL) and chrysophanol (25.8, 51.6, 129, 258, 516, 1290 and 2580 ng/mL) spiked in drug-free blank plasma (IS 2.5 ng/mL) by HPLC.

Recovery. Drug-free plasma samples of rats were spiked with three different concentrations of aloe emodin (56.0, 280.0 and 1400.0 ng/mL) and chrysophanol (51.6, 258.0 and 1290.0 ng/mL). Then fixed amounts of IS were added to plasma for normalization. The plasma samples were processed according to the above-mentioned extraction method. The recovery was calculated using the internal standard method.

Precision and accuracy. Plasma samples with aloe emosin (at concentrations of 56, 280 and 1400 ng/mL) and chrysophanol (at concentrations of 51.6, 258 and 1290 ng/mL) were prepared. Intra- and inter-day assays were repetitively carried out on the plasma samples six different times within one day and for three different days, respectively. Analytical precision was evaluated by calculating the RSD of variances intra- and inter-day, while accuracy was assessed by expressing the mean calculated concentration as a percentage of the nominal concentration, reflecting the difference between the measured concentration and spiked concentration.

Pharmacokinetics analysis. The concentration–time data were computer fitted using the Pharmacokinetics 3p97 program edited by the Mathematics Pharmacological Committee, Chinese Pharmacological Society. The following pharmacokinetic parameters were calculated: peak concentration (C_{max}), time of maximum plasma concentration (T_{max}), half-life ($t_{1/2}$), area under the concentration–time curve ($AUC_{0–12\,h}$), clearance (Cl) and elimination rate constant (K_e). In addition, other parameters were also investigated. The retention time ($MRT_{0–12\,h}$) was calculated using the statistical moment method of non-compartmental pharmacokinetic analysis.

RESULTS AND DISCUSSION

Extraction and recovery

The method for simultaneous determination of aloe emodin and chrysophanol required only 0.2 mL plasma that was extracted using a simple diethyl ether extraction procedure. The mean recoveries of aloe emodin and chrysophanol from rat plasma were 92.3 and 93.7% with mean RSD of 2.1 and 1.9%, respectively (Table 1).

Chromatographic selectivity

Figure 2 shows the chromatograms obtained following the analysis of drug-free plasma, drug-free plasma spiked with aloe emodin, IS and chrysophanol, and plasma sample collected 30 min after oral administration of

Table 1. Recoveries of aloe emodin and chrysophanol in rats plasma ($n = 5$)

Compound	Concentration added (ng/mL)	Recovery[a] (mean ± SD, %)	Average (%)	RSD[b] (%)	Average (%)
Aloe emodin	56.0	89.9 ± 1.5	92.3	1.7	2.1
	280.0	92.7 ± 2.1	92.3	2.3	2.1
	1400.0	94.2 ± 2.3	92.3	2.4	2.1
Chrysophanol	51.6	93.2 ± 1.2	93.7	1.3	1.9
	258.0	95.1 ± 3.2	93.7	3.4	1.9
	1290.0	92.9 ± 2.6	93.7	1.1	1.9

DOI: 10.1002/bmc

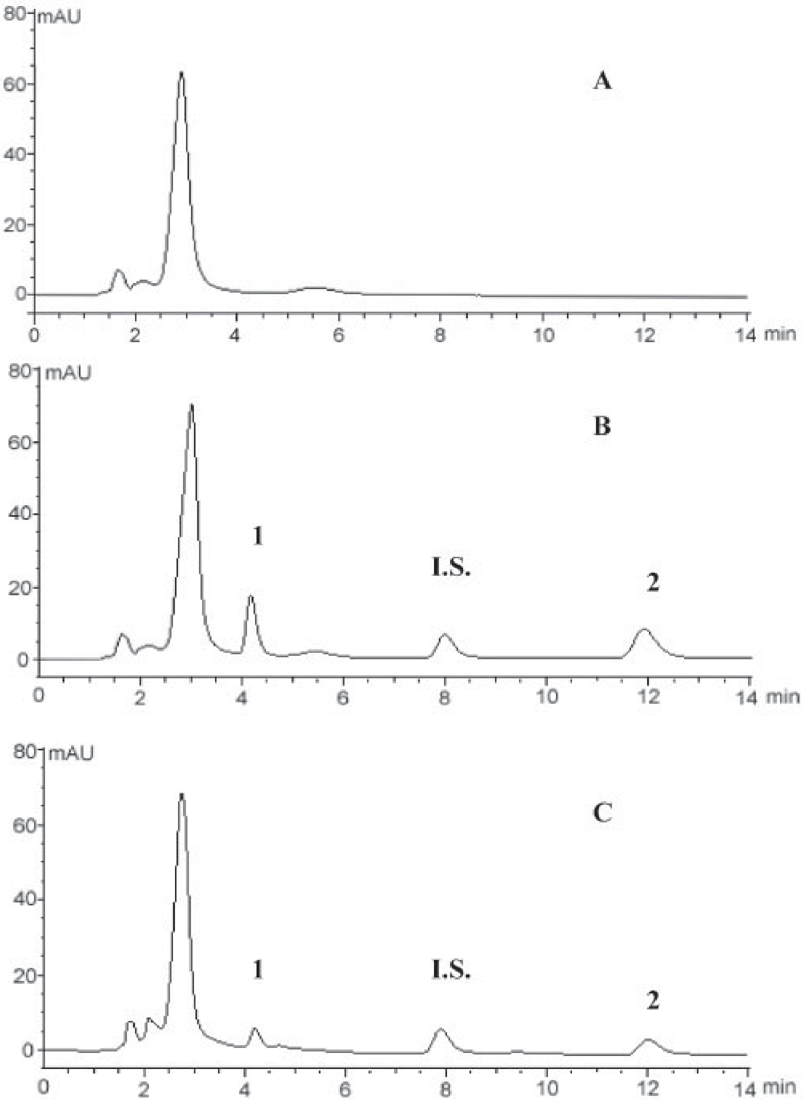

Figure 2. Typical chromatograms for the determination of aloe emodin and chrysophanol in plasma samples: (A) chromatogram of a blank plasma sample; (B) chromatogram of a plasma sample spiked with aloe emodin (**1**) and chrysophanol (**2**) and internal standard (IS); (C) chromatogram of theplasma sample collected at 1 h after oral administration of DCQD.

DCQD. Based on the chromatographic behavior of pure reference standards, peaks labeled 1 and 2 in Fig. 3 were identified as aloe emodin and chrysophanol with retention times of 4.24 and 12.17 min, and IS with a retention time of 7.98 min. Under the chromatographic conditions described, complete resolution was achieved and no interfering peaks were observed within the time frame in which aloe emodin, IS and chrysophanol were detected. The method used in this study is very rapid with a run time of 14 min for each analysis. This suggested efficient sample preparation and clean-up. The use of methanol and acetic acid as mobile phase resulted in baseline separation and satisfactory peaks.

Calibration curves

The calibration curve for aloe emodin was linear (r^2 = 0.9993) over the concentration range 28–2800 ng/mL; a regression equation of $y = 3522.417x - 5.549$ (where x is the peak area ratio and y the concentration of analyte) was obtained. Its detection limit based on a signal-to-noise ratio of 3 was 7 ng/mL and the quantitation limit was 14 ng/mL with an RSD of 11.10%. For chrysophanol it was linear (r^2 = 0.9991) in the range 25.8–2580 ng/mL; a regression equation of $y = 3245.867x + 27.395$ was obtained. Its detection limit was 6.45 ng/mL and quantitation limit was 14.6 ng/mL with an RSD of 14.6%. These ranges were found to be adequate

DOI: 10.1002/bmc

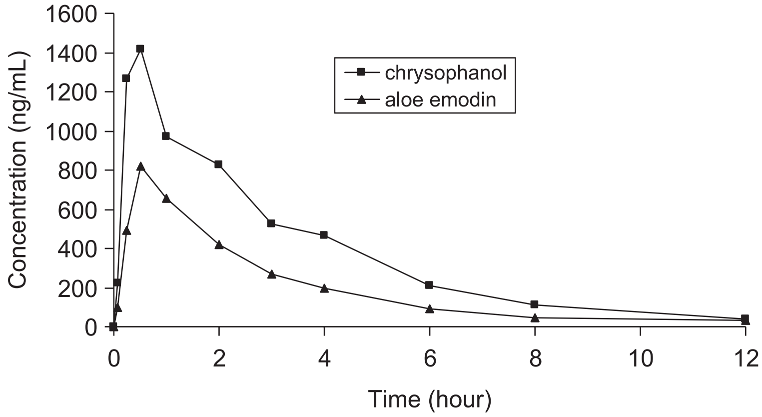

Figure 3. Plasma concentration–time curve of chrysophanol and aloe emodin after oral administration of DCQD (at a dose 10 g/kg body weight). Each point and bar represent the mean ± SD (n = 7).

Table 2. Intra-day variability for aloe emodin and chrysophanol (n = 6)

Compound	Added concentration (ng/mL)	Measured concentration (ng/mL, mean ± SD)	Accuracy[a] (%)	RSD[b] (%)
Aloe emodin	56.0	58.72 ± 1.95	104.8	3.3
	280.0	278.37 ± 5.21	99.4	1.8
	1400.0	1412.77 ± 23.25	100.9	1.6
Chrysophanol	51.6	53.24 ± 1.21	103.2	2.2
	258.0	252.17 ± 4.82	97.7	1.9
	1290.0	1292.59 ± 18.65	100.2	1.4

[a] Accuracy (%) = (mean of measured concentration/nominal concentration) × 100.
[b] RSD (%) (relative standard deviation) = (SD/mean) × 100.

Table 3. Inter-day variability for aloe emodin and chrysophanol (n = 6)

Compound	Nominal concentration (ng/mL)	Measured concentration (ng/mL, mean ± SD)	Accuracy[a] (%)	RSD[b] (%)
Aloe emodin	56.0	56.47 ± 1.32	100.8	2.3
	280.0	287.25 ± 8.28	102.6	2.8
	1400.0	1389.60 ± 32.13	99.2	2.3
Chrysophanol	51.6	51.14 ± 1.19	99.1	2.3
	258.0	260.37 ± 8.28	100.9	3.2
	1290.0	1305.26 ± 28.35	101.2	2.1

[a] Accuracy (%) = (mean of measured concentration/nominal concentration) × 100.
[b] RSD (%) (relative standard deviation) = (SD/mean) × 100.

for the concentrations observed from the analysis of collected plasma samples. It showed considerable linearity and had similar precision and accuracy to those reported in a study by Zaffaroni *et al*. (2003).

Precision and accuracy

The reproducibility of the method was defined by examining both intra- and inter-day variance. The intra- and inter-day precision assay also gave satisfactory results (Tables 2 and 3). The method was found to be highly precise, with an RSD of <3.5% and accuracy in the range 97.7–104.8% for each of the concentrations tested. The results showed that the method has good reproducibility.

Pharmacokinetic study

In the study, the established method successfully determined aloe emodin and chrysophanol in rats after oral administration of DCQD, and the mean estimated

DOI: 10.1002/bmc

Table 4. Pharmacokinetic parameters of aloe emodin and chrysophanol in rats

	Mean ± SD (n = 7)	
Parameters	Aloe emodin	Chrysophanol
T_{max} (h)	0.643 ± 0.244	0.393 ± 0.283
C_{max} (μg/L)	924.948 ± 341.711	1553.003 ± 449.685
$t_{1/2}$ (h)	2.222 ± 0.414	2.302 ± 0.490
K_e (1/h)	0.323 ± 0.067	0.314 ± 0.075
Cl/F (L/h/kg)	160.947 ± 46.679	230.365 ± 132.691
$AUC_{0-12\,h}$ (μg/L h)	1922.518 ± 631.673	4011.255 ± 582.545
$MRT_{0-12\,h}$ (h)	2.887 ± 0.257	3.174 ± 0.539

pharmacokinetic parameters of these two compounds are listed in Table 4. The time courses of the plasma aloe emodin and chrysophanol concentration showed that C_{max} was reached between 0.25 and 1 h after the oral administration of DCQD, and then gradually declined. The distributions of aloe emodin and chrysophanol in plasma and their elimination from plasma followed the one-compartment model. However, when aloe emodin was administered intraperitoneally (Takizawa *et al*., 2003), the T_{max} was between 1 and 2 h, and the half-life increased to 3.375 ± 0.335 h, which was substantially greater than the duration observed in this study. Therefore, the pharmacokinetic parameters vary greatly due to the difference in the mode of administration of medication. The time and time intervals at which DCQD is orally administered should be based on the pharmacokinetic parameters in our study.

Although the oral administration dose of DCQD contained 171 mg/kg aloe emodin and 55 mg/kg chrysophanol, the concentrations of aloe emodin in rat serum were much lower than those of chrysophanol. This implied that aloe emodin has a high binding activity to organs and a low absorption and blood distribution. Another possible cause was the low bioavailability of aloe emodin due to the fact that aloe emodin is metabolized into rhein *in vivo*. In contrast, the concentrations of chrysophanol were much higher than those of aloe emodin—even the dosage administered was approximately one-third that of aloe emodin. This indicated that chrysophanol was well absorbed and slow eliminated, even though it could be transformed to aloe emodin via methyl oxidation.

Although in this study aloe emodin was found in rat plasma after oral administration, it was not detected in humans after repeated administration of senna laxatives (Krumbiegel and Schulz, 1993). These differences might be attributed to the fact that different subjects were used for the experiments. First, the absorption, distribution, metabolism and excretion of aloe emodin were very different in different experimental subjects, while protein binding of aloe emodin was not measured in the study. It cannot be ruled out that the protein binding rate of aloe emodin is high in humans but low in rats. Second, the biotransformation of anthraquinones in different species and ethnic groups might differ because of the involvement of cytochrome P450 (Mueller *et al*., 1998). According to pharmacogenetics, the response to drugs shows significant inter-individual and inter-ethnic group differences due to differences in drug metabolism and responsiveness, genetic polymorphisms in genes encoding the drug metabolizing enzymes and drug receptors, and the interactive effect of environmental and genetic factors on drug-metabolizing enzymes (Zhou, 2003).

CONCLUSION

This is the first study to simultaneously determine aloe emodin and chrysophanol in rats plasma after oral administration of DCQD. The HPLC method shows considerable linearity and has similar precision and accuracy. The pharmacokinetic parameters showed that chrysophanol was better absorbed with higher concentrations in plasma than aloe emodin. They both eliminated slowly in male rats after a single oral dose of the DCQD. This method is very simple and rapid, requiring a simple diethyl ether extraction procedure and HPLC analyses with a run time of only 14 min. The assay is suitable for identifying the plasma and tissue levels of aloe emodin and chrysophanol in preclinical investigations, and the pharmacokinetic parameters may be used to guide the clinical application of traditional Chinese materia medica and prescriptions.

Acknowledgments

This project was supported by National Natural Science Foundation of China (30400576) and partly by (90409010). The authors would like to thank Ms Ning Qi-Zhi and Zhang Hong-Min for their kind help during the study and Zhao Yu for improving the language.

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DOI: 10.1002/bmc